

Final Independent Test Performance Comparison

Six Deep Learning Architectures | Independent Test Set: 2,399 Images | Primary Metric: Macro F1

Model	Base Acc (%)	Improved Acc (%)	Base Macro F1 (%)	Improved Macro F1 (%)	Base Bal. Acc (%)	Improved Bal. Acc (%)	Base MCC	Improved MCC	Base Loss	Improved Loss	Better
ResNet50	99.7082	99.8749	99.6439	99.8555	99.5815	99.8912	0.996685	0.998579	0.011052	0.005143	Improved
MobileNet	99.5415	99.6248	99.5014	99.5705	99.4812	99.5807	0.994788	0.995737	0.014917	0.012934	Improved
EfficientNet-B0	99.7499	99.7916	99.7310	99.7004	99.6897	99.7747	0.997157	0.997630	0.011990	0.010794	Baseline
ConvNeXt-Tiny	99.7916	99.9583	99.7331	99.9490	99.6667	99.9333	0.997633	0.999526	0.016694	0.004070	Improved
VGG16	98.3743	99.5832	98.1193	99.4848	98.1644	99.3507	0.981520	0.995262	0.068091	0.027355	Improved
InceptionNet	99.3331	99.7916	99.2636	99.7964	99.2238	99.8148	0.992424	0.997631	0.022385	0.011315	Improved

FINAL BEST MODEL: IMPROVED ConvNeXt-Tiny

Accuracy: 99.9583% | Macro F1: 99.9490% | Balanced Accuracy: 99.9333%
MCC: 0.999526 | Test Log Loss: 0.004070

Model selection followed the predeclared hierarchy: Macro F1 → Balanced Accuracy → Accuracy → Test Log Loss. Higher is better except Log Loss.